

Dev Prakash Jha

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Research Profile

Applied mathematician working at the interface of control theory, dynamical systems, partial differential equations, and scientific machine learning. My research focuses on observability and null controllability of parabolic systems with delay and memory, Carleman estimates, and controllability of degenerate and non-autonomous PDEs. I also investigate approximation, controllability, and robustness of semi-autonomous neural ordinary differential equations. My broader research interests concern the mathematical analysis of dynamical systems and the development of connections between control-theoretic methods and learning-based models.

Research Interests

- Scientific machine learning and neural ordinary differential equations
- Approximation, robustness, and controllability of nonlinear dynamical systems
- Control and observability of partial differential equations
- Delay and memory systems; degenerate and non-autonomous parabolic equations
- Numerical analysis and computational methods for evolution equations
- Fractional evolution equations and operator-theoretic methods

Education

Ph.D. in Mathematics, Indian Institute of Space Science and Technology (IIST), India
Expected completion: January 2027 | CGPA: 8.7/10
Supervisor: Prof. Raju K. George
Research area: control theory of PDEs, fractional evolution equations, delay systems, and controllability.

M.Sc. in Mathematics, Indian Institute of Technology Jodhpur (IIT Jodhpur), India
CGPA: 8.15/10; **Third topper**.

B.Sc. in Mathematics, Tilka Manjhi Bhagalpur University (TMBU), India
73%; **First topper**.

Selected Publications

1. Dev Prakash Jha and Raju K. George, "Observability and Null Controllability for Parabolic Systems with Delay and Memory." *SIAM Journal on Control and Optimization (SICON)*, accepted for publication, 2026.
2. Dev Prakash Jha and Raju K. George, " Memory Obstructs Null Controllability: Two Dichotomies for Linear Systems with Delay and Memory Effects." *International Journal of Systems Science*, accepted, 2026.
3. Dev Prakash Jha and Raju K. George, "Geometric Characterization of Memory-Type Null Controllability for Parabolic Equations with Moving Controls." *Mathematics of Control, Signals, and Systems*, Revision submitted, 2026.

4. Dev Prakash Jha and Raju K. George, "Existence and Uniqueness of Mild Solutions and Evolution Operators for a Class of Non-Autonomous Conformable Fractional Semi-Linear Systems and Their Exact Null Controllability." *International Journal of Dynamics and Control*, 2025. DOI: 10.1007/s40435-025-01844-x.

Preprints and Manuscripts

1. Dev Prakash Jha and Raju K. George, "Controlled Semiautonomous Neural Ordinary Differential Equations for Approximation and Controllability of Nonlinear Control-Affine Systems." Submitted to *IEEE Transactions on Neural Networks and Learning Systems*; under review.
2. Dev Prakash Jha and Raju K. George, "Turnpike-Preserving Universal Approximation for Controlled Semiautonomous Neural ODEs." Submitted to *IEEE Transactions on Automatic Control*; under review.
3. Dev Prakash Jha and Raju K. George, "Robust Controllability and Width–Robustness Tradeoffs for Semi-Autonomous Neural ODEs." Submitted to *Siam journal of control and optimization*.
4. Dev Prakash Jha, Akambadath K. Nandakumaran, and Raju K. George, "Memory-Type Null Controllability for Non-Autonomous Degenerate Parabolic Equations in Non-Divergence Form." Submitted to *Journal of Differential Equations*.
5. Dev Prakash Jha, Akambadath K. Nandakumaran, and Raju K. George, "Memory-Type Null Controllability for Non-Autonomous Degenerate Parabolic Equations in Divergence Form." Submitted to *Journal of Evolution Equations*.
6. Dev Prakash Jha and Raju K. George, "Approximate Controllability of Fractional Evolution Equations with Nonlocal Conditions via Operator Theory." arXiv:2501.17652, 2025. Submitted to *Fractional Calculus and Applied Analysis*; under review.
7. Dev Prakash Jha and Raju K. George, "An Operator-Theoretic Framework for the Optimal Control Problem of Nonlinear Caputo Fractional Systems." arXiv:2504.09611, 2025. Submitted to *Journal of Optimization Theory and Applications (JOTA)*; under review.

Research Experience and Projects

- Controllability and observability for parabolic systems with delay and memory.
- Carleman estimates and controllability with moving control regions.
- Null controllability of degenerate parabolic equations, including time-dependent diffusion and boundary degeneracy.
- Fractional evolution systems with nonlocal conditions and operator-theoretic analysis.
- Semi-autonomous neural ODEs: approximation and controllability of nonlinear control-affine systems; robustness and numerical evaluation.

Technical Skills

- Programming and scientific computing: Python, MATLAB, C/C++.
- Numerical methods: numerical simulation for dynamical systems and PDEs.
- Scientific writing: LaTeX.
- Machine learning: teaching-assistant experience in Foundations of Machine Learning and neural ODE research.

Teaching Experience

- Teaching Assistant, Foundations of Machine Learning (MA618).
- Teaching Assistant, Calculus (MA111).
- Teaching Assistant, Linear Algebra, Integral Transforms, and Calculus of Variations (MA221C).

Awards and Fellowships

- CSIR–UGC NET with Junior Research Fellowship (JRF), Mathematical Sciences, 2024; All India Rank 153.
- CSIR–UGC NET, Mathematical Sciences, 2021; All India Rank 51.
- GATE Mathematics, 2021; score 405.

- IIT-JAM Mathematics, 2018; All India Rank 350.
- Senior Research Fellowship; Institute Fellowship; Travel Grants.

Workshops and Schools

- Recent Developments in Applied and Computational Mathematics (RDACM), IIST, 2026.
- Recent Advances on Control Theory of PDE Systems (RACP), ICTS, 2024.
- NCM Workshop on Control Theory for Partial Differential Equations, IISER Thiruvananthapuram, 2023.
- 85th Annual Conference of the Indian Mathematical Society, IIT Kharagpur, November 2019.

Professional Memberships

Society for Industrial and Applied Mathematics (SIAM); American Mathematical Society (AMS).

References

Prof. Raju K. George (Ph.D. supervisor)
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Prof. Akambadath K. Nandakumaran (research collaborator)
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